The Substitution rules

5.5

GOALS:

- In this section, we will learn:

To substitute a new variable in place of an existing expression in a function, making integration easier.



- However, our antidifferentiation formulas don't tell us how to evaluate integrals such as

$$\int 2x\sqrt{1+x^2}\,dx$$

And to find this integral, we change from the variable x to a new variable u.

Example:

Find



$$\int 2x\sqrt{1+x^2}\,dx$$

$u = 1+x^2$ $\quad \frac{du}{dx} = \frac{2x}{1}$ $\quad dx = \frac{du}{2x}$

$$\int 2x\sqrt{u}\cdot\frac{du}{2x}$$

$$\int \sqrt{u}\cdot du \rightarrow \frac{u^{3/2}}{3/2} \quad \text{flip} \quad \frac{2}{3}\cdot(1+x^2)^{3/2}+C$$

The Substitution Rule:

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int \boldsymbol{F'(g(x))g'(x)dx} = \int \boldsymbol{F'(u)du}, \qquad \boldsymbol{where\ u = g(x)}$$

Remark:

$$\int F'\big(g(x)\big)g'(x)dx = F(g(x)) + C$$

because, by the Chain Rule,

$$\frac{d}{dx}\Big[F(g(x))\Big] = F'(g(x))g'(x)$$

Example:

$u = x^4 + 2$

$u' = 4x^3\,dx$



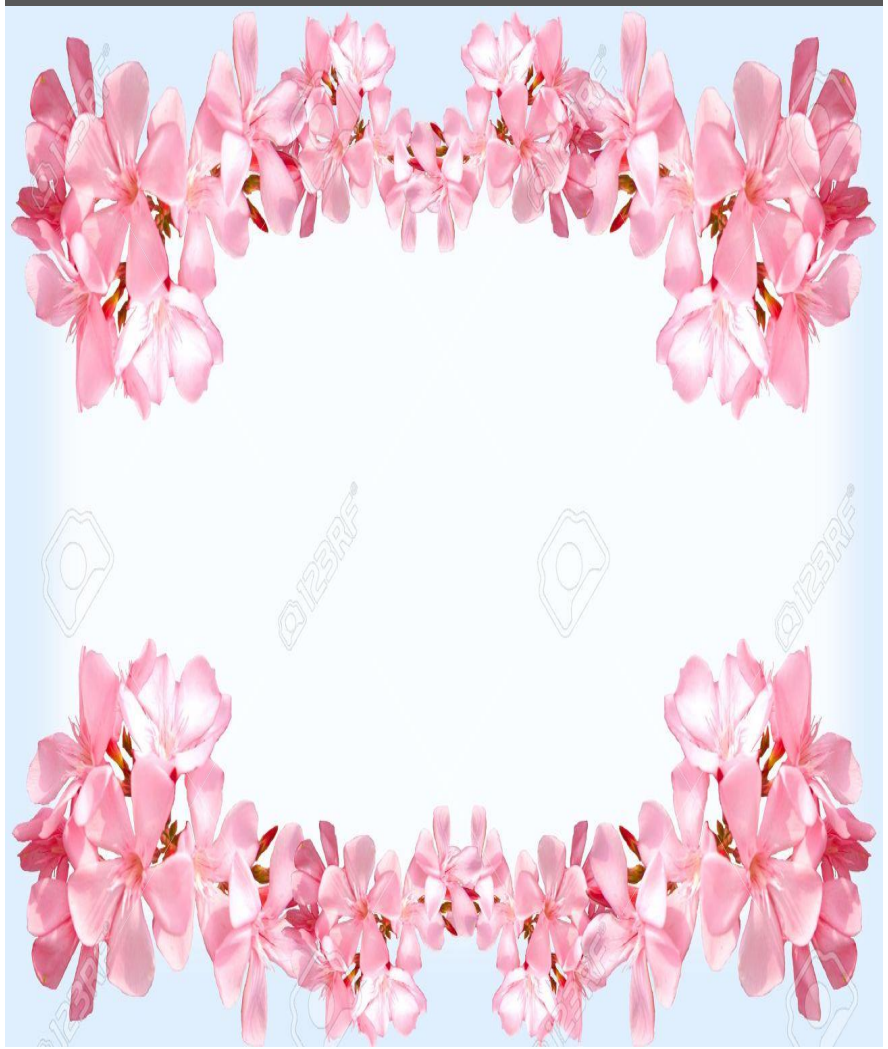
- **Find**

$$\int x^3 \cos(x^4 + 2)\, dx$$

Note:

- The main challenge in using the rule is
  to think of an appropriate substitution.

Example:



Find

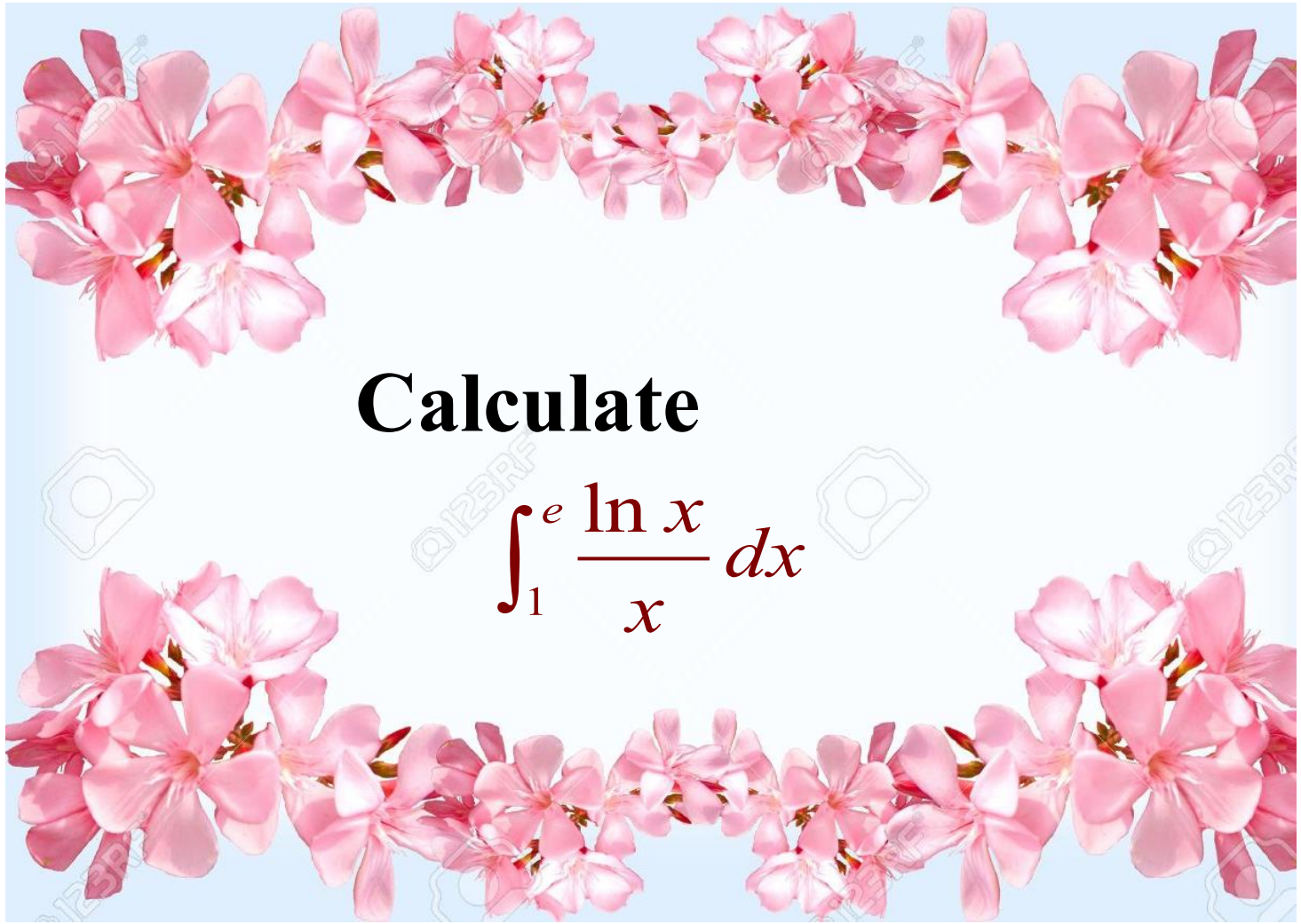
$$\int \tan x \, dx$$

$$\int \sqrt{2x+1} \, dx$$

$$\int \frac{x}{\sqrt{1-4x^2}} \, dx$$

$$\int \sqrt{1+x^2}\, x^5 \, dx$$

Example



Suppose *f* is continuous on [–*a* , *a*].

a.If *f* is even, [*f*(–*x*) = *f*(*x*)], then

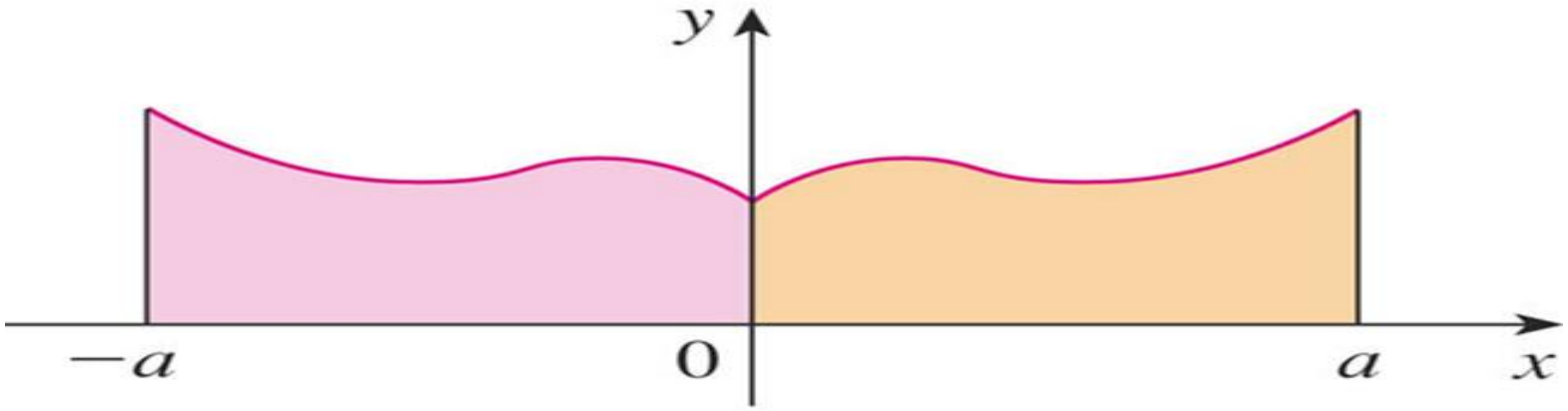
$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

b.If *f* is odd, [*f*(-*x*) = -*f*(*x*)], then

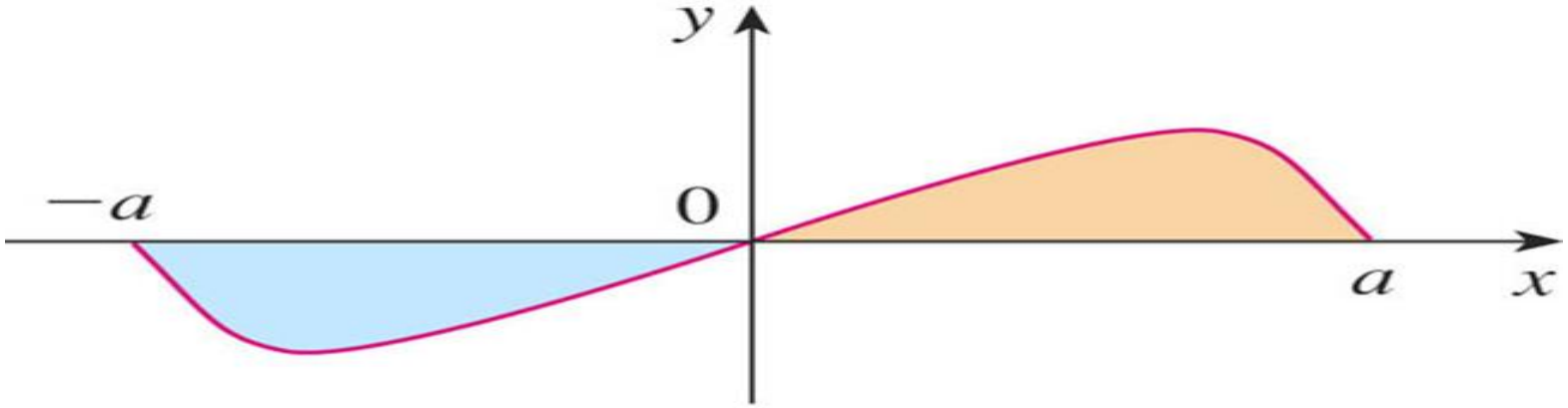
$$\int_{-a}^{a} f(x)\,dx = 0$$

INTEGS. OF SYMM. FUNCTIONS

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(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

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$$\int_{-2}^{2} (x^6 + 1)\, dx$$

$$\int_{-1}^{1} \frac{\tan x}{1 + x^2 + x^4}\, dx$$

Example: Find the following

Find

$$\int \frac{\tan x^{-1}}{x^2+1}\,dx$$

Find $\int \frac{dt}{\cos^2 t\sqrt{1+\tan t}}$

Example

Find $\int sint\, sec^2\, cost\, dt$